What is linked to failing Mathematics?

Findings from 395 secondary-school students in two Portuguese schools



Tarık Beyhan
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One in three students did not pass

We looked at school records and a questionnaire for 395 students (2005 to 2006) to see which characteristics go together with a lower final grade and with failing (below 10 out of 20).

33%

failed the final Mathematics grade (130 of 395)

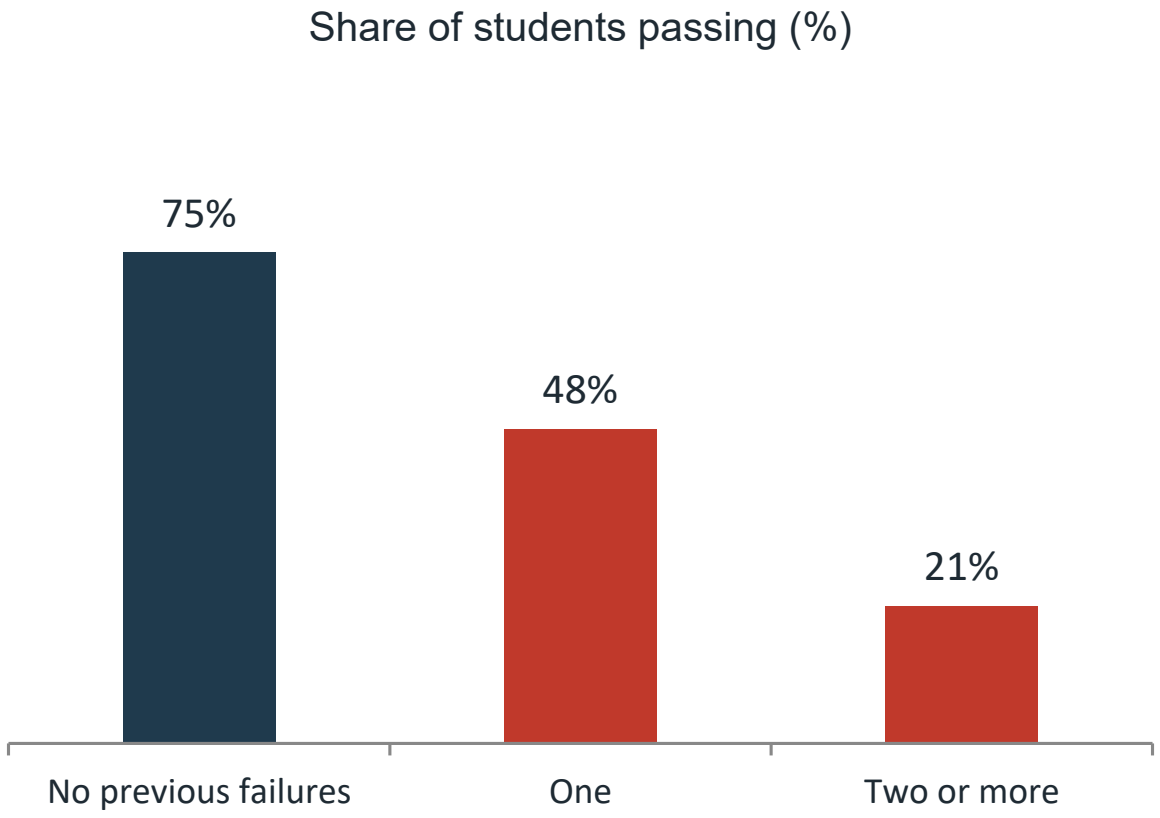
1 in 10

had a final grade of 0, most likely because they did not sit the exam

21%

pass rate among students who had already failed two or more classes

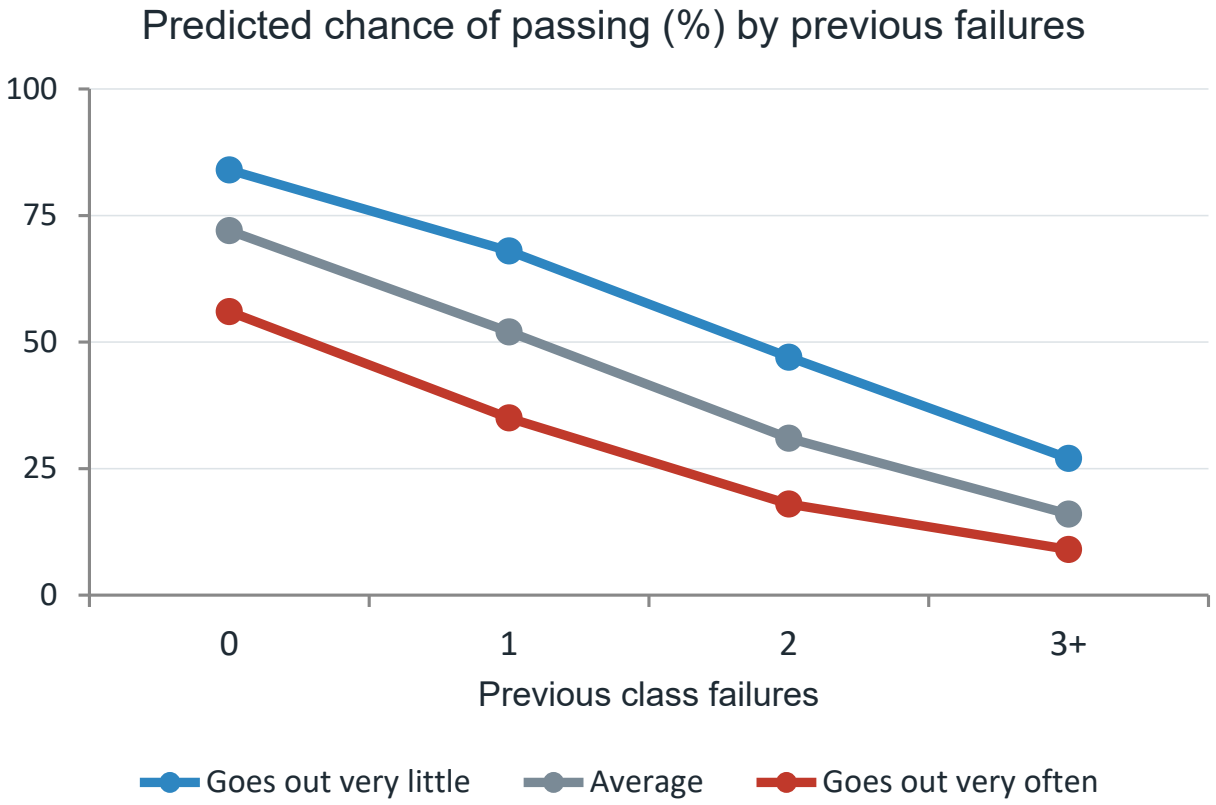
Past failures are the clearest warning sign



What we see

- Students who already failed one class pass at about half the rate of those with no failures; after two or more, at under a third.
- This holds after taking study time, family background, sex and absences into account.
- Going out with friends very often and lower maternal education show smaller, but consistent, links.

What this means for a typical student



72% → 31%

Predicted chance of passing for an otherwise typical student with no previous failures compared with two.

Practical use: past failures are already known to the school at the start of the year, so they are a simple, early flag for offering extra support.

How confident can we be?

1

Most of the story is missing

The characteristics we have explain only about one fifth of the differences in grades. Ability, teaching quality and home circumstances are not measured.

2

Not a reliable screening tool

The model correctly flags only about one in three students who actually fail. It should support, not replace, teachers' judgement.

3

Some results depend on the non-completers

When the 38 students without a final exam are set aside, more absences clearly go with lower grades, and the gap between boys and girls shrinks by about 40%.

4

Small, old, local sample

Two schools in Portugal, one year (2005 to 2006), one subject. Other schools and systems may differ.

What we can and cannot conclude

Supported by the data

- Students with past failures are at much higher risk of failing again.
- Frequent socialising and lower maternal education go with slightly lower grades.
- About 1 in 10 students did not complete the course; their records need checking.

Not supported

- That failing, going out or a mother's education causes lower grades.
- That boys are better at Mathematics, or that extra support does not work.
- That the model can decide which individual students will fail.